THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 25	$\lvert 30h - 5.5m \rvert$, smaller of the two.
2	**(a)** 1170	$360n + 90$.
3	**(a)** 180	Net turn.
4	**(a)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
5	**(a)** 7	The hour hand turns 0.5° per minute.
6	**(c)** 4	Each line gives one straight angle.
7	**(d)** 72	$360^\circ \div 5$.
8	**(d)** 150	Multiply the angle between neighbours.
9	**(b)** 105	$\lvert 30h - 5.5m \rvert$, smaller of the two.
10	**(a)** 1890	$360n + 90$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand $= 125^\circ$, minute hand $= 60^\circ$ (1); Smaller angle $= 65^\circ$ (1).	1+1
12	Total $= 360 \times 2 + 90 = 810^\circ$ (1); Right angles $= 9.00$ (about 9) (1).	1+1
13	Each pair of rays makes one angle (1); $\frac{8 \times 7}{2} = 28$ (1).	1+1
14	Hour hand $= 205^\circ$, minute hand $= 300^\circ$ (1); Smaller angle $= 95^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand at 55° (1); Minute hand at 300° (1); Smaller angle between them 115° (1).	1+1+1
16	Minutes $= 120 \div 6 = 20$ (1); Time is 2:20 (1); Angle between hands $= 50^\circ$ (1).	1+1+1
17	Slices $= 2 \times 2 = 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
	OR: Hour hand at 95° (1); Minute hand at 60° (1); Smaller angle between them 35° (1).	1+1+1
18	Minutes $= 180 \div 6 = 30$ (1); Time is 4:30 (1); Angle between hands $= 45^\circ$ (1).	1+1+1
	OR: Slices $= 2 \times 3 = 6$ (1); Angle $= 360 \div 6 = 60^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand $= 35^\circ$ (1); (b) Minute hand $= 60^\circ$ (1); (c) Difference $= 25^\circ$, smaller angle $= 25^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 180 - 45 + 90 = 225^\circ$ (1); Direction 225° from North (1); To face North again: turn clockwise 135° more (1); Type of the final turn: reflex (2).	1+1+1+2
20	(a) Hour hand $= 85^\circ$ (1); (b) Minute hand $= 300^\circ$ (1); (c) Difference $= 215^\circ$, smaller angle $= 145^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 270 - 90 + 135 = 315^\circ$ (1); Direction 315° from North (1); To face North again: turn clockwise 45° more (1); Type of the final turn: reflex (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 270	Net turn.
2	**(d)** 60	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
3	**(a)** 12.5	The hour hand turns 0.5° per minute.
4	**(a)** 5	Each line gives one straight angle.
5	**(b)** 60	$360^\circ \div 6$.
6	**(c)** 160	Multiply the angle between neighbours.
7	**(d)** 70	$\lvert 30h - 5.5m \rvert$, smaller of the two.
8	**(b)** 2610	$360n + 90$.
9	**(d)** 90	Net turn.
10	**(b)** 120	The hands are placed symmetrically about the 12-6 line, so the angle is the same.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total $= 360 \times 3 + 45 = 1125^\circ$ (1); Right angles $= 12.50$ (about 12) (1).	1+1
12	Each pair of rays makes one angle (1); $\frac{9 \times 8}{2} = 36$ (1).	1+1
13	Hour hand $= 215^\circ$, minute hand $= 60^\circ$ (1); Smaller angle $= 155^\circ$ (1).	1+1
14	Total $= 360 \times 4 + 180 = 1620^\circ$ (1); Right angles $= 18.00$ (about 18) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand at 155° (1); Minute hand at 60° (1); Smaller angle between them 95° (1).	1+1+1
16	Minutes $= 240 \div 6 = 40$ (1); Time is 5:40 (1); Angle between hands $= 70^\circ$ (1).	1+1+1
17	Slices $= 2 \times 4 = 8$ (1); Angle $= 360 \div 8 = 45^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand at 200° (1); Minute hand at 240° (1); Smaller angle between them 40° (1).	1+1+1
18	Minutes $= 300 \div 6 = 50$ (1); Time is 7:50 (1); Angle between hands $= 65^\circ$ (1).	1+1+1
	OR: Slices $= 2 \times 5 = 10$ (1); Angle $= 360 \div 10 = 36^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand $= 135^\circ$ (1); (b) Minute hand $= 180^\circ$ (1); (c) Difference $= 45^\circ$, smaller angle $= 45^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 135 - 225 + 270 = 180^\circ$ (1); Direction 180° from North (1); To face North again: turn clockwise 180° more (1); Type of the final turn: straight (2).	1+1+1+2
20	(a) Hour hand $= 185^\circ$ (1); (b) Minute hand $= 60^\circ$ (1); (c) Difference $= 125^\circ$, smaller angle $= 125^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 90 - 180 + 360 = 270^\circ$ (1); Direction 270° from North (1); To face North again: turn clockwise 90° more (1); Type of the final turn: reflex (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 25	The hour hand turns 0.5° per minute.
2	**(d)** 6	Each line gives one straight angle.
3	**(a)** 45	$360^\circ \div 8$.
4	**(a)** 135	Multiply the angle between neighbours.
5	**(b)** 175	$\lvert 30h - 5.5m \rvert$, smaller of the two.
6	**(b)** 3330	$360n + 90$.
7	**(a)** 135	Net turn.
8	**(a)** 150	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
9	**(d)** 45	The hour hand turns 0.5° per minute.
10	**(d)** 7	Each line gives one straight angle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each pair of rays makes one angle (1); $\frac{10\times9}{2} = 45$ (1).	1+1
12	Hour hand $= 380^\circ$, minute hand $= 240^\circ$ (1); Smaller angle $= 140^\circ$ (1).	1+1
13	Total $= 360 \times 5 + 270 = 2070^\circ$ (1); Right angles $= 23.00$ (about 23) (1).	1+1
14	Each pair of rays makes one angle (1); $\frac{12\times11}{2} = 66$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand at 295° (1); Minute hand at 300° (1); Smaller angle between them 5° (1).	1+1+1
16	Minutes $= 60 \div 6 = 10$ (1); Time is 8:10 (1); Angle between hands $= 175^\circ$ (1).	1+1+1
17	Slices $= 2 \times 6 = 12$ (1); Angle $= 360 \div 12 = 30^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand at 305° (1); Minute hand at 60° (1); Smaller angle between them 115° (1).	1+1+1
18	Minutes $= 150 \div 6 = 25$ (1); Time is 10:25 (1); Angle between hands $= 162.5^\circ$ (1).	1+1+1
	OR: Slices $= 2 \times 9 = 18$ (1); Angle $= 360 \div 18 = 20^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand $= 265^\circ$ (1); (b) Minute hand $= 300^\circ$ (1); (c) Difference $= 35^\circ$, smaller angle $= 35^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 225 - 45 + 180 = 360^\circ$ (1); Direction 0° from North (1); To face North again: turn clockwise 0° more (1); Type of the final turn: none (already facing North) (2).	1+1+1+2
20	(a) Hour hand $= 335^\circ$ (1); (b) Minute hand $= 60^\circ$ (1); (c) Difference $= 275^\circ$, smaller angle $= 85^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 315 - 135 + 90 = 270^\circ$ (1); Direction 270° from North (1); To face North again: turn clockwise 90° more (1); Type of the final turn: reflex (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.